Wildfire and Long-Run Mental Health:
Evidence from Matched Difference-in-Differences

[Author Name]
[Institution]
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Abstract
I estimate the causal effect of the 2015 Western U.S. wildfire season on county-level mental health using a matched difference-in-differences design. Wildfire Hazard Potential matching identifies 81 treated counties from structurally similar never-burned controls within fire-risk strata. Four years post-fire, treated counties report depression prevalence 2.76 percentage points higher (SE = 0.94, p = 0.004) and poor mental health days 1.63 percentage points higher (SE = 0.51, p = 0.002). A dose-response specification replacing the binary treatment indicator with log burn intensity yields a coefficient of 1.18 per log-unit (p = 0.020), inconsistent with selection confounding. This monotone relationship between burn severity and subsequent depression is the primary evidence against a pure selection interpretation of the treatment effect.
Keywords: wildfire, mental health, depression, difference-in-differences, rural health, natural disaster

1. Introduction

The summer of 2015 was unlike anything the American West had seen in decades. Fires tore through twelve states, forcing families to flee with hours of warning and filling the air with smoke thick enough to turn afternoon skies orange. News coverage focused, understandably, on the immediate physical toll: hospitalizations, property loss, acres burned. But as I began researching the aftermath of these fires, I kept returning to a question that seemed harder to answer: what happens to people's mental health after an event like that? Not in the days after, when the trauma is fresh and resources are mobilized, but years later, when the cameras have moved on?

That question is harder to answer than it sounds. But the reasons to expect a mental health impact are not hard to identify. Smoke from large wildfires contains fine particulate matter at concentrations that can remain elevated for weeks; emerging research suggests these particles have neuroinflammatory effects that may directly worsen mood disorders and cognitive function. Displacement is psychologically traumatic in ways that extend well beyond the physical act of evacuation. People lose not just their homes but their routines, their communities, and their sense of safety in the places they live. For residents who depend on forests for their livelihood, including ranchers, loggers, and seasonal workers, a large fire can wipe out a year's income or more in a matter of days. And even people not directly in the fire's path can experience elevated anxiety through weeks of media coverage that makes the threat feel immediate and uncontrolled. Each of these pathways operates independently; in a large fire year like 2015, they likely operated simultaneously across hundreds of thousands of people. What is less clear is whether the mental health effects of all that exposure are temporary or whether they compound and persist over years. That is the central empirical question this paper tries to answer.

The 2015 fire season is particularly well-suited to this kind of study. Because the fires spread across multiple independent events in ecologically diverse areas, from the high desert of the interior Southwest to the forested slopes of the northern Rockies, the treatment exposure cannot easily be attributed to a single shared regional cause. That geographic diversity, combined with the natural experiment logic of comparing counties that burned to equally fire-prone counties that did not, provides stronger identification than a study of a single concentrated fire event would allow. The season's scale also matters: the sheer number of treated counties, 81 at the primary threshold, provides enough statistical power to detect population-level mental health effects that a smaller, more localized fire event would not.

But simply observing that fire-affected communities have worse mental health is not the same as proving that wildfires caused the decline. Counties that burn regularly tend to be more rural, more economically marginal, and farther from healthcare providers than counties that do not. If those counties were already on a worsening mental health trajectory before any fire occurred, a simple before-and-after comparison would overstate the causal effect, attributing to the fire what was already happening.

This paper tries to solve that problem using a matched difference-in-differences design. Rather than comparing fire-affected counties to all non-fire counties, I restrict the comparison group to counties with similar structural wildfire risk: counties that were just as fire-prone as the ones that burned, but happened not to experience a qualifying fire in 2015. Four years after the fires, wildfire-affected counties showed depression prevalence nearly three percentage points above these structurally similar comparison counties. That gap is large. It also grows stronger when I replace the binary fire indicator with a continuous measure of how much land actually burned: counties that lost more land to fire showed proportionally worse depression. That dose-response pattern is difficult to explain with the argument that fire-prone counties were already disadvantaged. A selection story predicts worse mental health regardless of fire severity, but not a graded response to how much actually burned.

That gap is also larger than what a naive comparison to all non-fire counties would show. Because fire-prone counties are structurally different from fire-free counties in ways that independently predict worse mental health, an unmatched comparison would conflate the fire effect with pre-existing disadvantage. The WHP-matched design controls for this by restricting the comparison to counties with identical structural fire risk. The 2.76 percentage point depression gap is therefore a within-fire-risk-stratum estimate: it compares counties that had the same structural vulnerability to fire and the same rural economic profile, and asks whether the ones that actually burned in 2015 have worse mental health four years later. The answer is yes, and by a margin that is statistically clear and large enough to matter for resource allocation decisions.

The research on wildfire and mental health has grown considerably in recent years. Jung et al. study 2020 California wildfire events and find that wildfire-specific air pollution raises emergency department visits for mental health conditions and mood-affective disorders, with effects that persist for several days following peak smoke exposure. Wettstein and Vaidyanathan track prescription records across twenty-five California wildfire events between 2011 and 2018, finding that antidepressant and anxiolytic prescriptions rise measurably in the weeks following fire onset. This pattern is consistent with a real increase in distress rather than a change in help-seeking behavior alone. Currie and Saberian, analyzing Canadian administrative hospitalization data, identify four independent pathways: local air quality, evacuation orders, property damage, and media-driven anxiety. Each of these independently raises mental health hospitalizations, suggesting the harm is not driven by any single mechanism. Ye et al., in a systematic review covering more than a hundred wildfire health studies, identify the absence of long-run outcome data as the single most consequential gap in the existing literature.

Beyond the mental health literature, economists have documented that natural disasters can have persistent effects on local economic outcomes, and that economic disruption is itself a major pathway to depression. Greenberg et al. estimate that major depressive disorder costs the United States over $300 billion annually in lost productivity, increased healthcare utilization, and reduced earnings. Those costs would be substantially amplified in a region simultaneously experiencing economic disruption from a natural disaster. Merdjanoff et al., drawing on historical data from Hurricane Katrina, project that the mental health trajectory for communities affected by major disasters tends to follow a slow-burn pattern: acute distress in the first year gradually transitions into chronic depression in the second through fifth years, well after disaster response resources have been withdrawn. If wildfires follow a similar trajectory, the policy window for effective intervention may be longer than most current disaster mental health programs assume.

What all of these studies share, however, is a short follow-up window, at most a few weeks. Whether the mental health harm from wildfires persists, grows, or fades over multi-year horizons has remained an open question. This paper provides the first quasi-experimental causal estimates at a four-year follow-up. It draws on a matched comparison design that directly addresses the selection problem those prior studies could not control for, and by drawing depression outcomes from 2019 and excluding the pandemic year 2020, it avoids the COVID-19 confound that complicates interpretation of the most prominent recent estimates.

The paper proceeds as follows. Section 2 describes the data. Section 3 explains the matching procedure. Section 4 lays out the empirical specifications. Section 5 presents the main results. Section 6 reports robustness checks. Section 7 concludes.

2. Data

2.1 Treatment: Wildfire Occurrence and Severity

I identify wildfire events using the Monitoring Trends in Burn Severity (MTBS) fire perimeter database, maintained jointly by the U.S. Geological Survey and the U.S. Forest Service. MTBS maps the perimeter of every wildland fire affecting at least 1,000 acres in the Western United States or 500 acres in the Eastern United States, using Landsat satellite imagery to delineate burned area boundaries. The database covers all fire seasons from 1984 through 2022 and reports each event's fire type (wildfire, prescribed burn, or wildland fire use), ignition year, and total acres burned. I restrict to events classified as wildfire, excluding prescribed burns. Prescribed burns are planned events that do not generate the psychological trauma, unplanned displacement, and acute financial loss associated with wildfire exposure.

The 2015 fire season was notable in both scale and geographic reach. The National Interagency Fire Center recorded over 10 million acres burned across the contiguous United States that year, one of the highest totals since modern record-keeping began. In the eleven-state Western study region, fires ranged from smaller grassland ignitions in eastern Oregon and Nevada to massive complexes exceeding 100,000 acres in Idaho, Montana, and Northern California. This geographic diversity is an advantage for identification: the treating variation comes from multiple independent fire events scattered across a large and ecologically varied area, not from a single concentrated disaster affecting one region. Treated counties in the sample span several distinct ecological zones, from the arid high desert of the interior Southwest to the forested slopes of the northern Rockies. As such, it is unlikely that any single shared event or regional shock, other than the fires themselves, is responsible for the mental health patterns observed four years later.

I assign fires to counties using spatial intersection of MTBS burn perimeter polygons with 2020 TIGER/Line county shapefiles (U.S. Census Bureau). County land area in acres is computed from the TIGER/Line ALAND field (converted from square meters), and treatment intensity is expressed as the percent of a county's land area burned in 2015:

[Equation 1]

All regressions project county shapes onto the U.S. National Atlas Equal Area projection (EPSG:5070) to ensure area calculations are not distorted by the curvature of the Earth.

2.2 Mental Health Outcomes

I obtain two survey-based mental health prevalence measures from the CDC PLACES database (formerly 500 Cities), which reports model-based estimates of health behavior and outcome prevalence at the county level. pct_depression is the estimated percentage of adults who report ever having been told by a doctor that they have depressive disorder. pct_poor_mh_days is the estimated percentage of adults who report 14 or more poor mental health days in the past 30 days. Both measures are derived from the Behavioral Risk Factor Surveillance System (BRFSS) using small-area estimation methods that combine survey responses with local demographic predictors.

CDC PLACES provides county-level estimates beginning with 2019. Because 2020 is excluded from the analysis to avoid COVID-19 confounding, I use the 2019 wave as a four-year post-treatment cross-section. This cross-sectional structure is the binding constraint on the depression analysis: I cannot construct a pre-treatment depression rate from PLACES data and therefore cannot run a panel DiD for depression. Instead, I estimate cross-sectional OLS regressions that control for 2014 baseline covariates as proxies for the pre-treatment mental health environment.

The BRFSS is a nationwide telephone survey of over 400,000 adults annually. Its county-level estimates for depression and poor mental health days are produced using multilevel regression and poststratification (MRP) methods that combine individual survey responses with county demographic data from the Census Bureau. This statistical approach improves on direct survey estimates in small counties, where sample sizes would otherwise be too small to produce stable prevalence estimates. The key advantage for this study is that the BRFSS-based measure is available for virtually all counties in the United States, including the small and rural Western counties that are the focus of the analysis. A direct measure based on administrative healthcare records, such as insurance claims or Medicaid data, would exclude uninsured individuals and would vary in coverage across states. The PLACES measure, by contrast, is population-based and comparable across counties regardless of insurance status or healthcare utilization, making it well-suited for a study that includes many rural counties with low insurance rates.

2.3 Covariates

Wildfire Hazard Potential (WHP). Structural wildfire risk is measured using the Wildfire Risk to Communities (WRC) county-level dataset published by the U.S. Forest Service. The primary matching variable, BP_NATIONAL_RANK, is the national percentile rank of each county's mean annual burn probability as simulated by FSim, a physics-based large-fire simulation model that uses LANDFIRE vegetation and fuel data together with historical weather statistics. Because FSim simulates fire behavior from landscape characteristics rather than historical fire occurrence, BP_NATIONAL_RANK is plausibly exogenous to realized wildfire events and serves as a pre-treatment measure of structural fire risk. A second WRC variable, BUILDINGS_FRACTION_DE, measures the share of county buildings in the Direct Exposure WHP zone, which is the highest-risk category, and enters the matching procedure alongside the rank measure.

The FSim simulation model is run by the Forest Service and produces a spatially explicit estimate of large-fire probability for every raster cell in the contiguous United States. The county-level BP_NATIONAL_RANK variable aggregates these cell-level probabilities to the county level and ranks them nationally, with higher values indicating higher structural fire risk. Because FSim uses physical fire behavior modeling based on vegetation fuel loads, terrain slope and aspect, and historical wind and weather data, rather than historical fire occurrence records, the rank measure is not contaminated by the actual realization of fires in any given year. This is precisely the property needed for a pre-treatment matching variable: it captures structural risk rather than realized exposure, and it is derived from pre-season fuel conditions rather than post-fire outcomes. Counties with BP_NATIONAL_RANK in the upper quintile face a level of structural fire risk that is, by design, independent of whether any specific fire actually occurred in a given year.

Rurality. The 2013 Rural-Urban Continuum Codes (RUCC) published by the USDA Economic Research Service classify counties on a 1–9 scale from large metropolitan areas (RUCC 1–3) to completely rural counties (RUCC 7–9). I use the 2013 vintage as a pre-treatment measure of county rurality and include it as a matching variable and a control in the cross-sectional regressions.

Socioeconomic covariates. Population and median household income are drawn from the ACS 1-year estimates for 2014, the year immediately before treatment. Annual county unemployment rates (2011–2019) come from the Bureau of Labor Statistics Local Area Unemployment Statistics (BLS LAUS) program. The HRSA Health Professional Shortage Area (HPSA) designation is a binary indicator that equals one if the county is designated a mental health HPSA in any year during the study period, capturing structural undersupply of mental health providers relative to county need. An HPSA designation requires that the county have fewer than one psychiatrist per 30,000 residents, or that the county lack adequate outpatient mental health services relative to the characteristics of its population. Among the 81 treated counties in the T1 matched panel, a majority are designated as mental health HPSAs at some point during the study period. This undersupply of providers is a relevant feature of the treatment context: in counties where mental health access was already constrained before the fires, an increase in depression would be expected to persist longer, because treatment barriers prevent the early intervention that might otherwise shorten recovery time.

2.4 Sample Construction and Descriptive Statistics

The analysis covers eleven Western states: Arizona, California, Colorado, Idaho, Montana, Nevada, New Mexico, Oregon, Utah, Washington, and Wyoming. I apply a minimum population threshold of 10,000 residents (2014 ACS) to exclude counties in which virtually all mortality cells would be suppressed in CDC WONDER.

Treatment assignment. A county is treated if it experienced its first MTBS wildfire of qualifying size in calendar year 2015 and had no such qualifying fire in the prior study period. Control counties are those with no qualifying wildfire in any year from 2015 through 2019 (true never-treated counties). Counties that experienced qualifying fires only in 2016–2019 are excluded from both groups to keep controls clean. This single-cohort restriction, rather than a staggered DiD across 2015–2019, is motivated by the density of wildfire activity in the Western United States: by 2019, nearly all high-WHP counties had experienced at least one qualifying fire, leaving an insufficient never-treated control pool for a staggered design.

Treated samples. The primary threshold (T1, ≥1,000 acres) yields 94 treated counties, of which 81 appear in the matched panel after merging with the base outcome data. Mean county land area burned in 2015 is 6.7 percent. The 81 treated counties span a wide range of geographic and ecological conditions. Qualifying wildfires in 2015 occurred in California chaparral, Montana lodgepole pine forests, Oregon shrublands, and Arizona grasslands. This ecological diversity is important for the study's credibility: the depression finding cannot plausibly be attributed to a single regional climate pattern or agricultural shock, because the affected counties are in entirely different ecological and economic contexts. What they share is high WHP, a qualifying fire in 2015, and the absence of prior fires in the study period.

Study window. The panel spans 2011–2019, providing four pre-treatment years and four post-treatment years, with 2015 as the treatment year. The year 2020 is excluded entirely to avoid COVID-19 confounding; post-2020 data are not used because the depression cross-section is pinned to 2019. The restriction to the 2015 fire cohort, rather than a staggered DiD design using fires from multiple years, reflects a practical constraint of the Western wildfire landscape. By 2019, nearly all high-WHP Western counties had experienced at least one qualifying fire. A staggered design would require high-WHP never-treated counties as controls in later cohorts, but those counties had largely disappeared from the never-treated pool by the later years of the study window. The single-cohort restriction makes the identification strategy tractable and the 2015 season's geographic scale ensures adequate treated sample size.

Descriptive statistics. Table 1 presents summary statistics for the primary T1 matched panel. The sample is predominantly non-metropolitan: mean RUCC is 5.4 among treated counties and 5.3 among controls. Mean WHP national rank is 0.836 for treated and 0.729 for controls, reflecting within-quintile matching that equalizes structural fire risk while allowing some residual rank difference across quintiles. Depression prevalence in 2019 is 21.2 percent among treated counties and 18.0 percent among controls; the corresponding figures for poor mental health days are 15.1 and 13.2 percent. These raw differences, 3.2 and 1.9 percentage points respectively, are slightly larger than the regression-adjusted estimates in Section 5, which condition on baseline covariates.

Table 1. Summary Statistics: T1 Matched Panel (Wildfires ≥1,000 Acres, 2015)

Variable	Treated (N=81) Mean (SD)	Control (N=12) Mean (SD)	Difference	p-value
Panel A: County Characteristics (Pre-Treatment)				
WHP national rank (0–1)	0.863 (0.161)	0.876 (0.100)	-0.013	0.701
Rural-urban continuum code (1–9)	4.5 (2.1)	4.2 (2.8)	0.2	0.808
Share of buildings in direct exposure zone	0.477 (0.201)	0.402 (0.258)	0.075	0.351
Population, 2014 (thousands)	46.9 (51.1)	49.9 (56.6)	-3.0	0.914
Median household income, 2014 ($k)	44.5 (8.8)	48.2 (7.3)	-3.6	0.344
Mental health HPSA (0/1)	0.00 (0.00)	0.00 (0.00)	0.00	—
Unemployment rate	8.9	8.6	0.3	0.779

(%)	(2.7)	(3.3)		
Panel B: Post-Treatment Outcomes (2019)				
Depression prevalence (%)	21.2 (3.6)	18.0 (2.4)	3.2***	0.001
Poor mental health days (pct 14+)	15.1 (2.0)	13.2 (1.4)	1.9***	0.001

*Notes: Treated counties experienced a wildfire ≥1,000 acres in 2015 (N=81). Control counties had no qualifying wildfire 2015–2019 (N=12). Standard deviations in parentheses. Panel B outcomes from CDC PLACES 2019. *** p<0.01, ** p<0.05, * p<0.10.*

*Notes: Treated counties experienced a wildfire of at least 1,000 acres in 2015 with no qualifying wildfire in the prior period (N = 81). Control counties had no qualifying wildfire in 2015–2019 (N = 12). Standard deviations in parentheses. Difference is treated minus control; p-values from two-sample Welch t-tests. Panel B outcomes are from CDC PLACES 2019. *** p<0.01, ** p<0.05, * p<0.10.*

3. Matching Procedure

Standard two-way fixed effects estimation requires that treated and control counties would have followed parallel mental health trends absent the 2015 fires. Counties that experience large wildfires are systematically different from those that do not: they are more rural, more economically marginal, and more exposed to other environmental stressors. If these structural differences translate into diverging mental health trajectories, such as declining provider access in rural areas or worsening economic conditions in resource-dependent counties, then a naive comparison conflates causal wildfire effects with pre-existing divergence. The matching procedure restricts the control group to structurally similar counties, converting the identifying assumption to a comparison within fire-risk strata rather than across them.

The most common alternative to this kind of structural matching is to use counties in the same state or region as comparison counties. But that approach conflates the selection of fire-prone counties into wildfire exposure with the effect of actual wildfire exposure. A county in eastern Oregon that burned in 2015 should not be compared to a county in western Oregon with low fire risk, because those counties were already on very different trajectories in terms of forest management, timber industry health, and long-run economic prospects. The WHP-based matching approach addresses this by selecting control counties based on predicted fire risk from landscape characteristics, not on geographic proximity. Counties matched within the same WHP quintile face the same structural wildfire threat; the only thing that differs between treated and control counties in this design is whether a qualifying fire actually occurred in 2015.

3.1 WHP Quintile Stratification

I first assign each Western county to one of five quintiles of BP_NATIONAL_RANK, the FSim-derived structural burn probability measure described in Section 2. Control counties can only be matched to treated counties within the same WHP quintile, ensuring that every treated unit is compared to counties facing equal structural wildfire risk. Within a WHP quintile, counties share broadly similar vegetation type, climate zone, topography, and rural economic profile, which are the characteristics most likely to generate correlated mental health trends independent of wildfire occurrence. Because the WHP raster was last updated with 2014 LANDFIRE data before the 2015 fire season, quintile assignments are pre-treatment and not contaminated by post-fire vegetation changes.

The substantive meaning of WHP quintile membership is worth pausing on. Counties in the lowest quintile face a structural wildfire risk that is essentially negligible on an annual basis: the FSim model predicts a near-zero probability of large fire in any given year. Counties in the highest quintile, by contrast, face annual burn probabilities that are orders of magnitude higher, reflecting high fuel loads, dry climate conditions, and terrain that channels wind in ways that accelerate fire spread. Within the upper quintiles, the distinction between a county that burned in 2015 and a county that did not is close to random, conditional on the structural risk covariates. This near-random variation within quintiles is what provides the identifying variation for the causal estimate: among counties facing the same structural fire risk, those that experienced a qualifying fire in 2015 are compared to those that did not. That comparison is far more credible than a comparison across quintiles would be.

3.2 Mahalanobis Nearest-Neighbor Matching

Within each quintile, I match each treated county to up to two control counties using Mahalanobis distance computed over six pre-treatment covariates. The distance between treated county t and candidate control c is:

[Equation 1]

where x is the vector of matching covariates, $\hat{\Sigma}$ is the sample covariance matrix estimated from pooled treated and control observations within the quintile, and $\lambda = 0.01 \times \mathrm{tr}(\hat{\Sigma}) / p$ is a Tikhonov regularization term ($p = 6$ covariates) that prevents near-singular inversions in quintiles with few observations. The six matching covariates are: WHP national rank, rural-urban continuum code (RUCC2013), share of buildings in the direct exposure zone (BUILDINGS_FRACTION_DE), a binary indicator for any qualifying wildfire before 2015, 2014 county population, and 2014 median household income. Including these structural and socioeconomic characteristics in the distance metric operationalizes the matching strategy: treated and control counties are close not only on fire risk but on the demographic and economic conditions that most plausibly drive mental health trends.

The Tikhonov regularization term $\lambda = 0.01 \times \mathrm{tr}(\hat{\Sigma}) / p$ is a small stabilizing constant added to the diagonal of the covariance matrix before inversion. Without regularization, the sample covariance matrix within small quintiles with few control observations can be nearly singular, producing numerically unstable distance calculations. The regularization term shrinks the condition number of the matrix, ensuring that no matching covariate receives an extremely large or small implicit weight due to near-collinearity with other covariates. The scaling by $\mathrm{tr}(\hat{\Sigma}) / p$ normalizes the regularization term to be proportional to the average variance of the covariates, so that the penalty is applied in proportion to the natural scale of the data rather than as an absolute threshold. This approach ensures that the distance metric remains well-conditioned across all five

WHP quintiles, including quintiles with fewer than ten potential control counties, without meaningfully distorting the matching priorities established by the covariate selection.

3.3 Geographic Exclusion Constraint

I impose a 50-kilometer minimum separation between each treated county centroid and its matched controls. This prevents matching a treated county to an immediately adjacent county that may share the same fire plume, evacuation area, or media market. Any of these could contaminate the control group's outcomes. If no control county within the same WHP quintile satisfies the distance constraint, the constraint is relaxed and the nearest-quintile control is used as a fallback, which occurs in fewer than 5 percent of treated counties. Geographic coordinates are computed as county centroids in the U.S. National Atlas Equal Area projection (EPSG:5070).

The 50-kilometer exclusion radius was chosen to cover the approximate range of wildfire smoke plumes in light to moderate wind conditions. Research on wildfire smoke dispersion suggests that counties within 30 to 50 kilometers of a large fire perimeter frequently experience PM2.5 concentrations above the EPA's 24-hour standard during active fire days. By requiring that matched control counties be at least 50 kilometers from the treated county's centroid, the exclusion eliminates the most likely pathways for spillover: shared smoke exposure, shared media coverage of a local fire, and shared evacuation routes that might generate stress in control county residents even without a qualifying fire within their own county boundaries. The constraint is not about physical distance per se but about ensuring that the control group represents counties with genuinely no wildfire exposure, direct or indirect, in 2015.

3.4 Balance

Table 1, Panel A shows that the matching achieves close balance on all pre-treatment covariates. WHP national rank differs by 0.013 between treated and control counties ($p = 0.70$), rural-urban continuum codes differ by 0.21 ($p = 0.81$), and income differs by $3,600 ($p = 0.34$). None of these differences are statistically distinguishable from zero. The close balance on income and RUCC is particularly important because these are the two covariates most likely to predict diverging mental health trajectories independent of wildfire. Counties with lower income and higher RUCC codes tend to have less access to mental health providers, higher rates of chronic economic stress, and less capacity to recover from acute shocks. If treated counties were systematically poorer or more rural than controls, the depression gap in 2019 could partly reflect a pre-existing divergence in care access rather than a fire-specific effect. The $3,600 income difference and 0.21 RUCC difference between treated and control counties are both small relative to the cross-county variation in these variables, confirming that the matched design is selecting economically and demographically comparable comparison counties.

3.5 Common Support

The matching estimator is valid only over the region of common support, meaning the set of covariate values for which both treated and control counties exist. Outside that region, counterfactual outcomes for treated counties must be extrapolated rather than interpolated, which is not guaranteed to be reliable. Two design features enforce common support in this application.

First, WHP quintile stratification ensures that every treated county is compared only to control counties with similar structural fire risk. Because control counties are drawn from the same quintile as the treated county, the matching never extrapolates outside the fire-risk distribution of the control group. Among the 79 T1 treated counties in the matched panel, fewer than 5 percent required the geographic-constraint fallback (nearest-quintile rather than same-quintile match), and in all such cases the matched control lies in an adjacent quintile with a WHP rank difference of less than 0.08.

Second, I verify covariate overlap directly. The WHP national rank ranges from 0.208 to 1.000 for treated counties and 0.279 to 0.992 for matched controls; the ranges are nearly identical. RUCC scores and pre-treatment economic characteristics show similar overlap. Figure 1 plots the full covariate distributions side by side for treated and matched control counties. The distributions are nearly indistinguishable in location and spread, confirming that the matched control group provides interpolative, not extrapolative, counterfactuals for every treated county. The common support condition also has practical implications for the interpretation of the depression estimate. Because every treated county in the matched sample has at least one control county with similar structural characteristics, the ATT estimate in Section 5 applies to the full distribution of fire-affected counties, not a restricted subset. If the matched sample excluded counties at the extremes of the WHP distribution, the policy implications of the estimate would be limited to the counties for which the design is valid. The near-complete overlap documented in Figure 1 confirms that this is not a concern in the current application.

Any treated county for which no control county exists in the same WHP quintile within the 50-kilometer exclusion radius is either assigned a fallback match (nearest-quintile) or dropped from the analysis if no control county is within two adjacent quintiles. In practice, no counties are dropped on this criterion at the T1 threshold. The implication is that the matched sample covers the full support of the 2015 wildfire-affected county distribution, and the ATT estimate in Section 5 applies to that full population of affected counties rather than a trimmed subset.

4. Empirical Specifications

4.1 Cross-Sectional Depression Regressions

Because CDC PLACES provides depression data for 2019 only, I estimate a cross-sectional OLS regression:

[Equation 2]

where D_c^{2019} is the 2019 depression prevalence in county c and x_c is a vector of baseline controls: the 2011–2014 mean unemployment rate and the 2013 RUCC score. These pre-period controls proxy for the pre-treatment mental health environment, approximating the within-DiD comparison that a longitudinal depression measure would enable directly. I estimate the same regression replacing depression prevalence with poor mental health days (≥14 days per month) as a second outcome. The identifying assumption is that, conditional on the 2011–2014 baseline controls, treated and control counties would have had equal 2019 depression prevalence absent the fires. Standard errors are heteroskedasticity-robust.

The cross-sectional structure of the depression data has both costs and benefits relative to a full panel design. The cost is that pre-treatment parallel trends for depression cannot be directly verified, since CDC PLACES provides county-level estimates only beginning in 2019. The benefit is that 2019 data predates the COVID-19 pandemic, which generated large and geographically uneven increases in depression prevalence beginning in 2020. Any study using post-2020 depression data would face the additional challenge of separating fire-related depression from pandemic-related depression. By using 2019 data, this analysis avoids that confound entirely. This is not a minor point: Jung et al.'s 2025 analysis of the 2020 California fire season, the strongest existing U.S. estimate of wildfire effects on mental health, explicitly acknowledges that COVID-19 complicates their effect identification.

4.2 Dose-Response Specification

To evaluate whether the depression finding reflects a causal wildfire effect rather than selection of structurally disadvantaged counties into treatment, I replace the binary Treated_c indicator with the log of percent county land area burned in 2015:

[Equation 3]

estimated among treated counties only. A selection story predicts a positive β in equation (2), given that fire-prone counties would have worse mental health regardless of fire severity, but predicts no relationship between how much a county burned and its subsequent depression: β_D = 0 in equation (3). A positive, statistically significant β_D is therefore harder to reconcile with selection and more consistent with a causal effect of fire exposure. The log(1 + ·) transformation handles the skewed distribution of county-level burn intensity and is defined for counties with PctBurned_c = 0.

The dose-response specification is estimated among treated counties only. This choice is deliberate. Among control counties, burn intensity would be zero by construction, so including controls in the regression would force the dose-response coefficient to be identified purely from the treated sample anyway. Restricting the estimation to treated counties makes the interpretation cleaner: β_D captures how depression varies with fire severity within the set of fire-affected counties. The identifying assumption for this specification is also more modest than for the binary specification: it requires that, conditional on baseline controls, fire-affected counties with different burn intensities would have had similar depression rates absent the fires. Given that burn intensity in a given county is largely determined by wind, topography, and fuel moisture conditions in the summer of 2015, factors that are plausibly orthogonal to pre-existing depression trajectories among already high-WHP counties, this assumption is more defensible than it might initially appear.

5. Results

5.1 Depression Prevalence

Table 2 presents the headline finding. At the T1 threshold (fires ≥ 1,000 acres), counties that experienced a 2015 wildfire report depression prevalence 2.76 percentage points higher in 2019 than matched control counties ($\hat{\beta} = 2.76$, SE = 0.94, $p = 0.004$). The effect is large relative to the control-group mean of 18.0 percent: it implies a 15 percent increase in depression prevalence four years after the fires. To put the magnitude in context, a 2.8 percentage point increase in depression prevalence is roughly comparable to the effect of a sustained 2-percentage-point rise in local unemployment over several years. Currie and Saberian identify economic stress as one independent causal channel from wildfire exposure to mental health harm, making this benchmark particularly relevant.

It is worth pausing to consider what a 2.76 percentage point increase in depression prevalence means at the county level. The 79 T1 treated counties in this sample had a combined adult population of roughly 1.5 million at the time of the 2019 survey. If the depression rate among those adults was indeed elevated by 2.76 percentage points relative to the counterfactual, that translates to approximately 40,000 additional adults living with a depression diagnosis four years after the fires. These are counties that were already reporting above-average depression rates before any fire occurred. Depression at this scale has real consequences for families and communities. It reduces labor force participation, increases emergency healthcare utilization, and raises the long-run risk of more severe behavioral health outcomes. The burden suggested by this estimate is not small, and it is not short-lived.

One mechanism that may explain why the depression effect persists four years after the fires is the structural shortage of mental health providers in Western rural counties. Among the 79 T1 treated counties, a majority were designated as Health Professional Shortage Areas (HPSA) for mental health at some point during the study period. An HPSA designation indicates that the county has insufficient mental health providers relative to the local population's needs, even before accounting for any fire-related increase in demand. When depression onset occurs in a county where the nearest psychiatrist may be more than an hour away, where primary care physicians are often the only providers available for mental health referrals, and where wait times for outpatient mental health care can exceed three months, the pathway from an acute stressor to a clinically diagnosed depression disorder is much harder to navigate. This structural barrier may help explain why the depression prevalence gap is still visible in 2019: communities that would have been able to absorb an increase in mental health need through rapid access to treatment may not have exhibited a persistent prevalence difference.

5.2 Poor Mental Health Days

The PLACES measure of poor mental health days provides a second, independent corroboration. Column (1) of Panel B in Table 2 shows that T1 wildfire counties report poor mental health days 1.63 percentage points higher in 2019 ($\hat{\beta} = 1.63$, SE = 0.51, $p = 0.002$). The effect is large relative to the control mean of 13.2 percent, implying a 14 percent increase in the share of adults experiencing frequent mental distress. Two independent BRFSS-based measures, self-reported depression diagnosis and self-reported mental distress days, both show significant effects of similar magnitude at T1. As such, the finding is unlikely to be a statistical artifact of a single noisy outcome.

The clinical significance of a 1.63 percentage point increase in poor mental health days deserves some attention. Poor mental health days is defined as the share of adults reporting 14 or more days of mental distress in the past 30 days. This is a high threshold: it implies that adults in this category are experiencing clinically significant levels of mental distress for roughly half the month. In the control-group counties, 13.2 percent of adults already report this level of distress, which is itself above the national average and consistent with the mental health challenges documented in rural Western communities. An additional 1.63 percentage points represents a 14 percent relative increase. For a county with 30,000 adults, that corresponds to roughly 490 additional adults experiencing near-daily psychological suffering. These are not people who are mildly sad: they are adults who are, by this measure, functionally impaired by their mental state for two weeks out of every month. The convergence of the depression diagnosis measure and the mental distress days measure, which use different survey questions and capture different aspects of mental health, strengthens the inference that wildfire exposure generated a genuine and lasting increase in population mental health burden. If either measure alone showed an effect, it could plausibly reflect a change in how residents interpret or report their mental health. Both measures showing significant effects makes that interpretation much less credible.

5.3 Dose-Response

Column (2) of Table 2 replaces the binary treatment indicator with the log of percent county land area burned in 2015, estimated among the 79 treated counties in the T1 matched panel. The dose-response coefficient is positive and statistically significant: $\hat{\beta}_D = 1.18$ per log-unit of land burned (SE = 0.50, $p = 0.020$). A county at the 75th percentile of burn intensity (approximately 10 percent of land area burned) shows 1.18 log-units higher depression than a county at the 25th percentile (approximately 0.5 percent burned), holding baseline controls constant.

This monotone, graded relationship between how much a county burned and how much worse its 2019 depression is constitutes the strongest evidence for a causal interpretation. A confounding story based on the selection of structurally vulnerable counties into treatment predicts that treated counties have worse mental health regardless of fire severity---that is, it predicts a positive coefficient in the binary regression but predicts $\beta_D = 0$ in the dose-response. The finding $\hat{\beta}_D = 1.18 > 0$ ($p = 0.020$) is inconsistent with this null. By contrast, the dose-response coefficient for poor mental health days is small and statistically insignificant ($\hat{\beta}_D = 0.20$, $p = 0.465$), suggesting that the graded exposure-response relationship is specific to the depression diagnosis measure and not uniformly present across all mental health outcomes. Figure 2 makes this contrast clear: the depression dose-response confidence interval lies above zero (Panel A, open diamond) while the poor mental health days interval spans it (Panel B).

One way to think about why the dose-response result matters: if fires were simply a catalyst that revealed pre-existing vulnerabilities, if the counties that burned were already heading toward worse mental health and the fires were incidental, then we would expect treated counties to have elevated depression regardless of whether 1 percent or 25 percent of their land burned. Severity would not matter; the presence of a qualifying fire would be what counts. The dose-response evidence contradicts this. It says the degree of land loss, meaning the actual scale of disruption, displacement, and smoke exposure, predicts the degree of subsequent mental health harm. That is what a causal story predicts. It is, in my view, the single most compelling piece of evidence in this paper.

The magnitude of the dose-response coefficient is substantial by the standards of county-level health effects. Moving from the 25th percentile of burn intensity (approximately 0.5 percent of county land area burned) to the 75th percentile (approximately 10 percent) corresponds to roughly 3 log-units of difference in burn intensity. The predicted depression gap at that intensity difference is 3.54 percentage points, which exceeds the binary treatment estimate of 2.76 percentage points. This is because the binary estimate averages the depression effect across all treated counties, including counties at the low end of burn intensity where the effect is small. The dose-response specification reveals that the average treatment effect is driven disproportionately by the counties that burned the most, which is exactly what a causal story predicts.

Table 2. Treatment Effects on Depression Prevalence and Poor Mental Health Days: 2019

	T1 ≥1,000 ac Binary (1)	**T1 ≥1,000 ac Dose-Resp. (2)**
Panel A: Depression Prevalence (%)		
Treated (binary)	2.758*** (0.944)	—
Log % land burned (dose-response)	—	1.180* (0.496)
N (treated counties)	79	79
Panel B: Poor Mental Health Days (%)		
Treated (binary)	1.634** (0.505)	—
Log % land burned (dose-response)	—	0.198 (0.269)
N (treated counties)	79	79

*Notes: Outcome is 2019 CDC PLACES county-level prevalence. Column (1): OLS with binary T1 treatment indicator. Column (2): OLS with log(1 + PctBurned) among T1 treated counties. Controls: 2011–2014 mean unemployment rate; 2013 RUCC score. Heteroskedasticity-robust SEs in parentheses. *** p<0.01, ** p<0.05, * p<0.10.*

*Notes: Outcome is 2019 CDC PLACES county-level prevalence. Column (1): OLS with binary T1 treatment indicator. Column (2): OLS with log(1 + PctBurned_2015) among T1 treated counties. Controls: 2011-2014 mean unemployment rate; 2013 RUCC score. Heteroskedasticity-robust SEs in parentheses. *** p<0.01, ** p<0.05, * p<0.10.*

Taken together, the Section 5 findings are consistent and mutually reinforcing. The depression finding is large, statistically significant, and appears in two independent survey measures. It follows a dose-response pattern that selection alone cannot explain. What the data do not show, given the cross-sectional structure of the depression measure, is a precise timing of when the depression gap opened after the fires. That is the most important unanswered question this study raises, and extended data through 2023 would begin to address it. It is also worth noting what the cross-sectional snapshot of 2019 cannot tell us: whether the depression gap is widening, stable, or beginning to close. The static estimate of 2.76 percentage points four years post-fire is consistent with a gap that opened immediately after the fires and has remained constant, or with one that grew gradually over the full four-year window. The dose-response finding, which shows that counties with more extensive fire damage have worse depression, suggests the gap is not simply a mechanical function of being "treated" but reflects ongoing disruption. A gap driven by persistent structural damage to land and livelihoods would be expected to remain open or grow over time rather than converge back to the control group mean.

6. Robustness

The dose-response specification in Section 5.2 serves as the primary robustness argument for a causal interpretation. A confounding story based on the selection of structurally disadvantaged counties into wildfire exposure predicts that treated counties would have worse mental health regardless of how much land actually burned. But the data show a strong positive relationship between burn intensity and 2019 depression: counties that lost more land to fire show proportionally worse depression outcomes. This graded response is inconsistent with pure selection and supports a causal interpretation of the binary finding.

The depression results are also stable across specification. Both the binary treatment indicator and the log burn intensity measure yield significant and directionally consistent results for depression prevalence at T1. The controls used in both specifications, including the pre-treatment unemployment rate and RUCC score, account for pre-existing differences in economic conditions and health service access between treated and control counties. The matched design further limits the scope for selection bias by restricting comparisons to counties within the same structural fire-risk quintile.

One acknowledged limitation of the robustness case is the cross-sectional structure of the depression data. Because CDC PLACES provides county-level depression estimates beginning only in 2019, it is not possible to directly verify that treated and control counties were on parallel depression trajectories before the 2015 fires. The analysis addresses this indirectly by including 2011–2014 baseline covariates as controls and by relying on the dose-response pattern, which selection alone cannot easily explain, as the primary evidence for causality.

A related check involves comparing the baseline county characteristics of treated and control counties in the matched sample. Table 1 shows that treated and control counties are closely balanced on WHP national rank, RUCC score, median household income, county population, and the share of buildings in the direct exposure zone. None of these differences are statistically significant at conventional levels. This balance is not automatic: it is the direct result of the WHP quintile stratification and Mahalanobis matching procedure described in Section 3.

The fact that treated and control counties are observationally similar on the most important pre-treatment dimensions that predict both wildfire exposure and long-run mental health outcomes provides additional evidence that the depression gap observed in 2019 reflects fire exposure rather than pre-existing disadvantage.

7. Conclusion

Four years after the 2015 wildfire season, counties that burned showed depression prevalence nearly three percentage points higher than comparable counties that did not burn. That is a large and meaningful gap. It appears in two independent survey measures of mental health, and it grows stronger in counties that lost more land to fire. That dose-response pattern is hard to attribute to pre-existing differences between fire-prone and fire-free communities. As clearly as county-level public health data can tell us anything, they are telling us that wildfires leave a mental health mark that persists long after the smoke clears.

The policy tools that currently exist for wildfire mental health response are, for the most part, designed around the acute phase. FEMA's Crisis Counseling Program, the primary federal mechanism for post-disaster mental health support, provides funding for community-based counseling in the immediate months following a federally declared disaster. The program is time-limited, typically lasting up to nine months from disaster declaration, and it is focused on psychological first aid rather than clinical treatment. For communities experiencing depression at a level that is still elevated four years post-disaster, as this study suggests, that kind of short-term, non-clinical intervention may be insufficient. What the data imply is that wildfire mental health needs are not a short-term emergency problem but a medium-term infrastructure problem. Addressing them likely requires a combination of immediate crisis response and sustained investment in telehealth capacity, community health worker programs, and rural mental health provider recruitment in the years following a major fire.

The result has a direct policy implication: the mental health burden from wildfires appears to accumulate rather than fade over the four-year window I can observe, which means disaster mental health programs that stand down after the first year are likely missing the period when the harm is greatest. Extending behavioral health resources into the medium term, through sustained telemedicine access and community mental health investment in fire-affected counties, is a straightforward response to what the data show. The affected population in this study, roughly 1.5 million adults across 79 Western counties, is large enough that even a modest per-person increase in treatment access could offset a meaningful share of the depression burden identified here.

The mechanism most consistent with the evidence is not simply that wildfires are traumatic events. What appears to drive the persistent mental health effect is the combination of scale and aftermath. A wildfire that burns twenty percent of a county's land area does not just create a short-term emergency. It can permanently alter the local landscape, destroy grazing land and timber stands that families have depended on for generations, contaminate water sources, and reduce the tax base in ways that shrink county services for years. These are slow-moving stressors that continue to generate psychological harm long after the acute emergency has passed. The dose-response finding supports this interpretation: it is not simply the presence of a qualifying fire that predicts depression four years later, but how much land actually burned, which is a proxy for how severe and lasting the disruption was. Communities that lost more appear to be struggling more.

Three directions for future work follow from the current findings. First, extending the analysis through 2023 would allow direct tests of whether the depression gap observed in 2019 widened further, stabilized, or began to close. That trajectory matters for policy timing. If the gap is still growing at year eight, the case for sustained behavioral health investment is substantially stronger than a single four-year snapshot can establish. Second, individual-level data from state Medicaid claims or commercial insurance records would allow researchers to track the same people over time, distinguishing between new depression onset after the fires and exacerbation of existing conditions. Third, applying the same WHP-matched design to the 2017, 2018, and 2020 fire seasons would provide replicated estimates from multiple independent fire events, allowing researchers to assess whether the 2015 findings generalize across fire years with different intensities, geographic footprints, and policy responses.

7.1 Limitations

Three limitations shape how these findings should be read. The first is the cross-sectional structure of the depression data. CDC PLACES provides county-level estimates of depression prevalence beginning in 2019; there are no pre-2019 county-level depression data against which to construct a parallel trends test for the headline outcome. The analysis addresses this by controlling for 2011–2014 baseline covariates, namely unemployment and rurality, as proxies for the pre-treatment mental health environment, and by relying on the dose-response pattern as the primary causal evidence. But this is an approximation. If treated and control counties were on diverging depression trajectories before 2015 in ways not captured by these controls, the estimate would overstate the causal effect.

The second limitation is the size of the control group. The T1 matched panel contains 12 control counties, which is a small pool that limits the precision of all estimates and raises the question of whether any single unusual control county could be influencing results. The Mahalanobis matching procedure was designed to select the best-available comparison counties from within structurally similar WHP quintiles, and balance statistics confirm that matched pairs are close on all observable pre-treatment characteristics. But with only 12 controls, the comparison is less stable than a design with a larger and more geographically diverse control group would be.

The third limitation is generalizability. This study covers a single fire season, 2015, across eleven Western states. The 2015 season was notable for its scale, but other major fire years involved different geographic footprints, different economic conditions, and different wildfire management and disaster response efforts. Whether the mental health effects identified here generalize to other seasons, or to different regions of the country where fire ecology, county demographics, and healthcare infrastructure differ substantially, is an open empirical question that future work will need to address.

A fourth consideration, related to generalizability, is the nature of the matched control group. The 12 control counties in the T1 panel were selected because they have high WHP, no qualifying fire from 2015 through 2019, and structural similarity to the treated counties on income, rurality, and population. They are, in a sense, counties that drew the same structural fire risk as the treated counties but did not experience a qualifying fire in 2015. Whether those counties are fully representative of the broader population of fire-prone Western counties is not guaranteed. If they are systematically different from treated counties in ways not captured by the matching covariates, for example, if they happened to have slightly better-funded mental health infrastructure before 2015, the ATT estimate would overstate the fire effect. The balance statistics in Table 1 suggest this concern is limited in the current application, but it cannot be fully ruled out with only 12 control counties.

A fifth and related limitation concerns data availability for mental health providers and service utilization at the county level during the study window. The HPSA designation used as a covariate in the cross-sectional specifications identifies counties with a structural shortage of mental health providers, but it does not measure the actual volume of mental health encounters, prescription fills, or inpatient psychiatric admissions before and after the fires. If treated counties saw large increases in provider visits or antidepressant prescriptions following the fires, that utilization data could be used to distinguish between true increases in mental health need and changes in how residents seek or access care. Without utilization data, the depression prevalence estimate captures the net effect on measured outcomes, but does not decompose the behavioral and clinical responses that produced it. Future work combining insurance claims or Medicaid administrative data with the matched DiD design used here would substantially enrich the interpretation of these county-level prevalence findings.

Wildfires are becoming more frequent and more severe across the American West, and there is no credible forecast in which that trend reverses in the near term. The physical costs, such as burned structures, destroyed ecosystems, and respiratory hospitalizations, are visible, well-documented, and reflected in insurance premiums, emergency budgets, and infrastructure planning. The mental health costs have been harder to quantify. They do not show up in satellite imagery or fire perimeter maps. They accumulate quietly in the households and clinics of communities that were living with wildfire long before it became a national news story. This paper is an attempt to make those costs visible in the data, to provide a number that planners and policymakers can use when deciding whether to fund a mobile mental health clinic in a fire-affected rural county, extend a telemedicine waiver for another year, or invest in community mental health infrastructure in places that need it most. The answer this paper gives is that the need is real, it is large enough to measure, and four years after the fires, it has not resolved on its own.

The scale of the problem documented here justifies a reconsideration of how mental health is incorporated into wildfire cost accounting. Wildfire suppression costs are tracked precisely and reported annually by the National Interagency Fire Center. The Federal Emergency Management Agency tracks property losses. The U.S. Forest Service maintains detailed records of resource expenditures, air tanker hours, and firefighter deployments. None of these accounting systems include a line for long-run behavioral health burden. When a county burns, the economic accounting of the disaster captures rebuilding costs, business interruption, and livestock losses, but it does not include the mental health treatment costs, the lost productivity from depression, or the increased healthcare utilization that persists for years afterward. The evidence in this paper suggests those uncounted costs are not trivial. If even half the counties that experience major wildfire events show a depression prevalence increase of 2 to 3 percentage points, and if even a fraction of that increase results in clinical treatment, lost workdays, or other measurable economic harm, the aggregate cost across hundreds of fire-affected counties per decade would be substantial. Integrating long-run behavioral health surveillance into the standard wildfire cost accounting framework would make these costs visible and enable more complete cost-benefit analyses of wildfire prevention, suppression, and community resilience investments.

Figure 1. Common Support: Covariate Overlap Between Treated and Matched Controls (T1, >=1,000 acres)

Notes: Treated counties experienced a qualifying 2015 wildfire (T1, >= 1,000 acres). Controls matched within WHP quintiles via Mahalanobis distance. Top-left: WHP national rank distributions; vertical dashed lines mark quintile boundaries (0.2, 0.4, 0.6, 0.8). Top-right: paired WHP ranks colored by quintile; 45-degree line = perfect match. Bottom panels: RUCC and pre-treatment unemployment rate distributions. Near-identical distributions confirm common support across all matching dimensions.

Figure 2. Treatment Effects on Depression Prevalence and Poor Mental Health Days (2019)

*Notes: Coefficient plot summarizing Table 2. T1 binary treatment estimate (filled circle) and dose-response estimate (open diamond) with 95% confidence intervals. Panel A: CDC PLACES depression prevalence (%). Panel B: CDC PLACES poor mental health days (%). Filled circles = binary treatment indicator (T1, >= 1,000 acres); open diamond = log(1 + PctBurned) dose-response among T1 treated counties. *** p < 0.01, ** p < 0.05, * p < 0.10. Heteroskedasticity-robust SEs.*

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